

Psycho-social factors and coping strategies as predictors of chronic evolution and quality of life in patients with low back pain: A prospective study

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Abstract

Low back pain (LBP) is a major problem of public health. Chronic pain is the most difficult to treat and the most expensive. The way patients cope with their pain may influence its outcome.

Aim: To identify coping strategies of LBP patients, and their influence on LBP evolution.

Methods: Ninety nine patients were assessed just after an acute LBP episode and one year later. Assessment tools included medical and social reports, scales of anxiety, depression, quality of life, locus of control (LOC), social support and coping strategies.

Results: One year after the initial episode, 67% of patients have improved and 33% had a chronic pain. A principal components analysis showed that two main dimensions might be identified inside the outcome: functional and emotional non-adjustment. Functional non-adjustment was predicted by male gender, reduction of activity, and history of trauma over one year. Emotional non-adjustment was only predicted by trait-depression. Reactions to pain were structured in four factors: distraction-praying, helplessness–hopelessness, cognitive restructuring and perceived control. Two of these factors predicted adjustment one year later: distraction-praying had a direct effect on functional non-adjustment, and helplessness–hopelessness on emotional issue.

Conclusions: Besides somatic factors, psychosocial predictors of LBP chronic evolution may be identified. Both aspects must be taken into account in order to prevent chronic pain. Perhaps cognitive-behavior therapy may help LBP patients to cope with pain in a better way.

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Keywords: Low back pain; Chronic low back pain; Psychological factors; Coping; Quality of life

1. Introduction

With a cumulative incidence as high as 70%, low back pain (LBP) represents an important factor of disabling

chronic pain and low quality of life in the adult population, and a devastating problem of Public Health because of its tremendous medical and social costs. Time to evolution plays a major role: acute LBP improves in 70–90% of the cases (Andersson, 1999) and appears relatively easy to deal with, whereas chronic LBP (CLBP) is at the same time the most important source of disability and expenses, and the most difficult to treat. The

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4–7% of CLBP patients who do not return to work within six months account for 75% of the medical and social costs (Maetzel and Li, 2002). So preventing a chronic evolution of LBP appears to be of great importance. This needs to identify when and why a LBP becomes chronic. Temporal definitions vary in the literature. Some authors, wishing to put the stress on the importance of an early treatment, are recommending to look at LBP as chronic as soon as one month and a half (Atlas, 1996). But in most of recent reviews and consensus reports acute LBP is defined by a pain duration within three months, and CLBP by a pain duration over three months (Bigos et al., 1994; Duquesnoy, 1994; Clinical guidelines for the management of acute low back pain, 1996; Guidelines for the management of back-injured employees, 1993).

From a clinical point of view, LBP is a complex, multifactorial process which depends on constitutional, somatic, psychological and environmental factors as well, so that the classical medical approach which deals mainly with somatic factors often fails to prevent a chronic evolution. With regard to psychological and sociological factors, many works demonstrated that they are of minor importance as far as acute LBP is concerned, but play a major role in CLBP (Truchon and Fillion, 2000). In a recent and exhaustive review of the literature, Linton concluded that “psychosocial variables generally have more impact than biomedical or biomechanical factors on back pain disability, and are clearly linked to the transition from acute to chronic pain disability (Level A evidence)” (Linton, 2000). Some works put the stress on sociological factors, showing that job dissatisfaction, low incomes and leisure are found associated with high levels of CLBP (Pope, 1989; Symonds et al., 1996; Vällfors, 1985). Among psychological factors, anxiety and depression have been for a long time the best identified of these factors (Clauw et al., 1999; Epping-Jordan et al., 1998; Fisher and Johnston, 1998; Kessler et al., 1996; Pincus et al., 2002; Polatin et al., 1993; Schermelleh-Engel et al., 1997). Cognitive factors such as fear-avoidance beliefs and passive coping are also related to pain and disability with a Level A evidence (Linton, 2000). Self-reported feelings of disability and irrational and/or negative beliefs about pain such as kinesiophobia and fear avoidance have been found associated with a chronic evolution (Fritz et al., 2001; Gatchel et al., 1995; George et al., 2003). Distraction, catastrophizing and praying/hoping are linked with negative aspects of CLPB (pain intensity, disability and poor emotional state) whereas cognitive restructuring and pain control are associated with positive evolution of CLBP (Dozois et al., 1996; Haythornthwaite et al., 1998; Kröner-Herwig et al., 1996; Lewandowski, 2004; Lin and Ward, 1996; McCracken et al., 1998; Riley et al., 1999; Robinson et al., 1997; Tuttle et al., 1991). However most of these studies are retro-

spective and/or cross-sectional, which prevents from analyzing precisely which psychodynamic process occurs when LBP becomes chronic. Moreover, asking retrospectively subjects who are feeling pain for a long time prevents them to provide reliable data about their psychological states and affects before the onset of pain (Koleck, 2001).

In the context of prospective studies, catastrophizing and fear-avoidance beliefs were found to be the most powerful predictors of pain and disability (Burton et al., 1995; Hasenbring et al., 1994; Klennerman et al., 1995; Linton and Hallden, 1998; Picavet et al., 2002). Passive coping (Potter and Jones, 1992), praying/hoping (Burton et al., 1995) and self-perception about the future development of pain and disability (Hazard et al., 1996; Linton and Hallden, 1998) were also found predictors of disability at a one-year-outcome.

But understanding how these variables interact with each others remain difficult, because of confounding factors, and most of these studies consider only pain criteria: chronic pain, onset of a new episode, or functional ones: disability, return to work, although quality of life and relationships between functional and emotional factors remain less documented. So prospective studies are needed to clarify the role of psycho-social factors in chronic both functional and emotional evolution of LBP.

The aim of the present study was to identify in an exploratory approach how patients deal with acute LBP, and how these reactions would expose to, or protect from, a chronic evolution and low quality of life.

2. Materials and methods

2.1. Study sample

Ninety-nine patients were enrolled consecutively and prospectively in the study when consulting a general practitioner (GP) for an acute LBP, and/or having stopped working or performing domestic activity because of pain within three months.

Exclusion criteria were:

- recurrent LBP or evidence of a previous LBP episode within two years,
- LBP not related to common, degenerative origin: inflammatory diseases, cancer, discal or spinal infection, Paget's disease, osteoporosis. . . ,
- recent spinal trauma ¹ (less than one week before),
- severe psychiatric disorders with hospitalization.

¹ Recent spinal trauma is a particular case, which involves specific coping strategies.

Table 1

Demographical, social and medical data on patients in sample and initial pain episode

Age from 18 to 60 year old; mean = 42.2; SD = 11.5
Gender: male 48; female 51
Marital status: married or similar: 71%; bachelors: 20%; divorced: 8%; widowed: 1%
Number of children: 0: 21%, 1: 19%, 2: 34%, 3: 19%, 4 and more: 7%
Obesity or excessive weight: 24%
Income: low: 12%, middle: 56%, high: 23%
History of spinal impairments: 31%, including: Scoliosis 13%, Scheuermann disease 2%, listhesis 2%, Spinal trauma: 14.1%
Previous LBP episode over 2 years: 53.3%
Sciatica: 20.2%, including 3% with surgery
Data on the initial LBP episode (T1): Evidence of discal pathology on CT-scan: 4%; facet joint pathology: 2%
Drug treatment: analgesics: 96.9%; anti-inflammatory drugs: 75.7%
LBP due to work-related accidents: 64.6%

Table 2

Evolution of the initial episode (T2)

Group	Evolution	Men	Women	Total
IP	Improvement	19	35	54 (60%)
	Improvement after relapse	4	2	6 (6.7%)
CLBPP	Chronic pain	16	7	23 (25.5%)
	Worsening chronic pain	6	1	7 (7.8%)
	Lost for follow-up	3	6	9

Ninety-nine patients fulfilling these criteria were recruited. They were 48 men and 51 women, (mean age = 42.2). They were interviewed by the first author between 10 and 90 days after the onset of LBP. Table 1 provides data about these patients, and about the initial LBP episode. One year later (Table 2), 54 patients related that they had improved and returned to work within 48.3 days on average after the first interview. Six other patients had a relapse during the year of follow-up, but were free of pain at one year. So these 60 patients were retrospectively classified as improved patients (IP) at the date of follow-up. Thirty patients were still suffering from LBP at this date, and thus were classified as chronic LBP patients (CLBPP). Last, nine patients were lost for follow-up.

3. Methods

3.1. Inclusion data

At the inclusion time (T1), the aims of the study were explained by the practitioner. Patients received written information for consent, and had a face-to-face interview to record demographic and medical data. Then questionnaires were administered in order to assess: (A) general psychological functioning and (B) transactional pro-

cesses. All questionnaires were validated, either previously, or for this study, using either principal components analysis (PCA) or structural confirmatory analysis.

(A) General Psychological Functioning:

- Trait form of State-Trait Anxiety Inventory (STAI, Spielberger et al., 1983) to assess general trends towards anxiety (validated by Bruchon-Schweitzer and Paulhan, 1993).
- Center for Epidemiologic Studies Depression-Scale (CES-D, Radloff, 1977), in a revised version developed by our group to assess trait-depression. A PCA performed on the answers of the 99 present patients revealed one factor (18 items, α of Cronbach = 0.89).
- Locus of Control Scale (LCS, Lumpkin, 1985) to assess the influence a person thinks he/she may have on what happens to him/her. This scale was validated by our group in a population of 347 adults including the 99 present patients: a PCA revealed one factor (6 items, α = 0.60).
- one question assessing job satisfaction of the patient.

(B) Transactional processes:

- State form of State-Trait Anxiety Inventory (STAI, Spielberger et al., 1983) to assess the state of anxiety during the interview (validated by Bruchon-Schweitzer and Paulhan, 1993).
- Center for Epidemiologic Studies Depression-Scale (CES-D) to assess the state-depression during the interview (Radloff, 1977; Fuhrer and Rouillon, 1989). A PCA performed on the answers of a sample of 125 LBP patients including the 99 present revealed one factor (18 items, α = 0.83).
- Cancer Locus of Control Scale (Pruyn et al., 1988), in a revised version we developed for chronic pain patients. A PCA performed on the answers of the 125 LBP patients revealed 3 factors: perceived control on pain course (CLCS-1, six items, α = 0.65), irrational beliefs about pain (CLCS-2, four items, α = 0.64), and internal causal attribution of pain onset (CLCS-3, five items, α = 0.67).
- Perceived Social Support Scale (PSSS): created for this study, this scale was derived from Social Support Questionnaire (Sarason et al., 1983), to assess patients' opinion on the social support they may hope for. A PCA performed on the answers of the 125 LBP patients revealed two factors: quality of social support (SSQ-1, four items, α = 0.86) and availability of social support (SSQ-2, four items, α = 0.63).
- Coping Strategies Questionnaire (CSQ, Rosenstiel and Keefe, 1983) to assess how patients are coping with pain. Structural confirmatory analyses

performed by our group in a population of 330 chronic pain sufferers proved the pertinence of a five-factor model: distraction (CSQ-1, five items), catastrophizing (CSQ-2, five items), reinterpreting pain sensations (CSQ-3, four items), ignoring pain sensations (CSQ-4, four items), praying (CSQ-5, three items). This structure converged with the one of Robinson et al. (1997) and Riley et al. (1999) but the coping self-statement factor was suppressed because its internal consistency was too weak.

- Visual Analog Scale (VAS) to assess pain intensity during the interview.

3.2. One-year follow-up data

- One year later, medical reports assessed how long patients have stopped working because of LBP, how many times they have consulted their general practitioner (GP) for LBP and GP's opinion on LBP evolution.²
- State form of STAI and CES-D were used again to assess the states of anxiety and depression.
- Two questionnaires assessed quality of life:
 - Medical Outcome Study short form General Health Survey (Stewart et al., 1988), a shortened form of the SF36 (Ware and Sherbourne, 1992), assessed functional limitations (MOS-1, 9 items, $\alpha = 0.90$), feeling of well-being (MOS-2, 4 items, $\alpha = 0.83$) and perceived good health (MOS-3, 3 items, $\alpha = 0.89$);
 - Nottingham Health Profile (Bucquet et al., 1990; Hunt and Wilkund, 1987) measured functional limitations (NHP-1, 13 items, $\alpha = 0.92$) and psychological distress (NHP-2, 14 items, $\alpha = 0.88$).

3.3. Data treatment

Data were anonymously recorded and statistical treatments were performed with the SPSS 11.0 software. Principal components analyses (PCA) were used to explore the structure of the reaction during acute LBP and the structure of criteria one year later.³ Hierarchical multiple regressions were performed to test the link between predispositional factors, transactional processes and criteria.

² LBP evolution was assessed from 'important improvement without relapse' to 'important aggravation after relapse'.

³ For factor analysis, we followed the recommendations of Fabrigar et al. (1999).

4. Results

4.1. Descriptive data

Global data from medical reports and psychological questionnaires are given on Table 3 for the two groups of patients (IP and CLBPP). Only few differences were found between the groups: scores significantly differed in causal attribution of pain onset, irrational beliefs, distraction, perceived good health, and of course number of GP visits pain, days on sick leave and functional limitations. Anxiety and depression levels as assessed by the STAI-state and the CES-D scores were similar in both groups at inclusion time T1, and were slightly lower at follow-up time T2 also in both groups, without any significant difference.

4.2. Outcome criteria

A PCA followed by oblique rotations was performed on data collected at T2 in order to identify the factorial structure of the outcome criteria. The scree-test (Cattell, 1966) was used for factor extraction: two factors explaining 70.8% of total variance (TV) were found. All of the 10 variables loaded more than |0.40| onto one of the two factors. The first one (37.5% of TV) was labeled Functional Non-Adjustment (FNA), since it included functional limitations (MOS-1 and NHP-1), duration of sick leave, number of consultations and GP's opinion, as opposed to the perceived health (MOS-3). The second factor (33.3% of TV) was labelled emotional non-adjustment (ENA), including severe psychological distress (NHP-2) with high levels of anxiety (STAI) and depression (CES-D) states, as opposed to feelings of well-being (MOS-2).

So, outcome and adjustment to LBP might be assessed with regard to these two factors, the functional one and the emotional one. These two dimensions were found significantly associated ($r = 0.25$; $p = 0.02$).

4.3. Transactional processes

Afterwards we tried to identify how LBP patients reacted to acute pain. A PCA was performed with Varimax rotations and scree-test (Cattell, 1966). Variables with loading below |0.40| or projecting on more than one axis (SSQ-1, CSQ-2 and VAS) were dropped out. Four factors were displayed (66.4% of TV). They were related to the following 10 variables (loading |0.40| or greater): anxiety and depression states, CLCS factors, SSQ-2, CSQ-1, CSQ-3, CSQ-4 and CSQ-5.

The first factor (20% of TV) was labeled distraction-praying (DP). It included one factor of the CLCS, irrational beliefs (CLCS-2), and two factors of the CSQ, praying (CSQ-5) and distraction (CSQ-1). The second factor (17.4% of TV), composed by high levels of anxiety

Table 3
Distribution of the main data in sample

	Improvement		Chronic pain	
	T1	T2	T1	T2
<i>N</i>	60		30	
Age	41.9(11.6)		42.8(11.2)	
<i>Predispositional factors</i>				
STAI-Y trait	40.2 (10.5)		37.7 (10.3)	
CESD - trait	30.2 (9.2)		27.3 (8.9)	
LCS	19.6(2.8)		18.3(3.4)	
<i>Transactional factors</i>				
STAI-Y state	43.9(13.7)	39.2 (14.9)	43.6(13.3)	40.0(15.1)
CESD state	33.6 (8.9)	30.8 (10.9)	32.9 (7.8)	29.1 (10.9)
CLCS-1, perceived control	17.5 (3.6)		17.8 (4.2)	
CLCS-2, irrational beliefs	8.3*(2.8)		10.2* (3.3)	
CLCS-3, causal attribution	12.6* (3.8)		9.1* (2.8)	
CSQ-1, distraction	12.6* (4.9)		15.7* (4.7)	
CSQ-2, catastrophizing	12.1 (4.7)		12.8 (4.7)	
CSQ-3, reinterpreting	6.5 (2.9)		7.1 (3.6)	
CSQ-4, ignoring	8.1 (3.1)		8.3 (3.9)	
CSQ-5, praying	5.6 (3.4)		6.1 (3.5)	
SSQ-1 quality of social support	20.3 (2.8)		21.2 (2.9)	
SSQ-2 social support availability	19.0(15.9)		16.6(18.7)	
VAS	8.5 (1.8)	2.3 (2.7)*	9.1(1.5)	4.9 (2.4)*
Evolution (T2)				
Days on sick leave		43.8* (84.1)		148.8* (134.8)
GP visits for pain, N		2.2* (3.9)		7.8* (7.9)
MOS-1, functional limitations		15.0* (5.0)		22.0* (4.8)
MOS-2, well-being		17.3 (4.8)		15.6 (5.0)
MOS-3, perceived health		10.5* (3.4)		6.6* (4.4)
NHP-1, functional limitations		16.6* (3.5)		21.3* (3.3)
NHP-2, psychological distress		17.8 (3.6)		19.0 (4.2)

T1 = Time of inclusion; T2 = one-year follow-up. STAI-Y = State-trait Anxiety Inventory; CES-D = Center for Epidemiologic Studies Depression Scale; LCS = Locus of Control Scale; CLCS = Cancer Locus of Control Scale; CSQ = Coping Strategies Questionnaire; VAS = Visual Analog Scale; MOS = Medical Outcome Study; NHP = Nottingham Health Profile. See text for details. SD in brackets.

* $p < 0.05$.

(SYAI) and depression (CES-D) was labelled helplessness–hopelessness (HH). The third factor (15.3% of TV) included two factors of the CSQ, reinterpreting (CSQ-3) and ignoring pain sensations (CSQ-4), was labelled cognitive restructuring (CR). The fourth factor (13.7% of TV) included two factors of the CLCS, perceived control on pain course (CLCS-1) and internal causal attribution of the onset of pain (CLCS-3), and one factor of the PSSS, availability of social support (SSQ-2), was named perceived control (PC).

4.4. Influence of medical, psychological and social factors on transactional processes

Further multiple regression analyses were performed on each of these four transactional processes (Table 4). Independent variables were all the predispositional factors (Fig. 1). DP was mostly used by men, with low income, low job satisfaction, who had no previous LBP experience and reduced their activity because of pain; these men were not usually anxious and presented an external LOC. People who were usually anxious and

depressive, with few children and low income, were mostly using HH. CR was used by men with a preserved life comfort during the acute pain. Finally, PC was mostly used by men with a high education, several children and a preserved life comfort.

4.5. Influence of predispositional factors and transactional processes on adjustment

The same method was applied to the two dimensions of outcome we observed at T2: functional and emotional non-adjustment (Fig. 1 and Table 5). Gender, activity, and history of trauma over one year were predictive of FNA: men with history of trauma over one year who reduced their activity because of pain were at risk of FNA one year after. ENA was only related to trait-depression. With regard to transactional processes, DP predicted a low functional adjustment, although PC seemed to influence good functional outcome, but data fell below significance. HH was predictive of a low emotional adjustment.

Table 4
Factors influencing transactional processes

Variables	<i>F</i>	<i>R</i> ²	Adjusted <i>R</i> ²	<i>β</i>
<i>Distraction-Praying</i>	7.565*	0.436	0.379	
Gender				−0.22*
Number of children				0.11
Job satisfaction				−0.20*
Income				−0.18*
Excessive weight				−0.13
Reduced activity				0.20*
Previous LBP episode				−0.21*
Trait-anxiety				−0.20*
Internal locus of control				−0.29*
<i>Helplessness–hopelessness</i>	12.727*	0.534	0.492	
Number of children				−0.17*
Job satisfaction				0.13
Income				−0.24*
Excessive weight				0.13
Reduced activity				0.11
Poor life comfort ^a				0.09
Trait-anxiety				0.24*
Trait-depression				0.42*
<i>Cognitive restructuration</i>	3.968*	0.146	0.109	
Gender				−0.24*
Income				−0.18
Previous LBP episode				−0.17
Poor life comfort				−0.28*
<i>Perceived control</i>	3.404*	0.209	0.148	
Gender				−0.33*
Education				0.32*
Number of children				0.25*
History of trauma over one year				−0.14
Previous LBP episode				0.13
Poor life comfort				−0.19*
Trait-anxiety				0.15

Multiple regression analyses.

^a This variable corresponded to a self-report of strong deterioration of life comfort due to lumbar pain.

* *p* < 0.05.

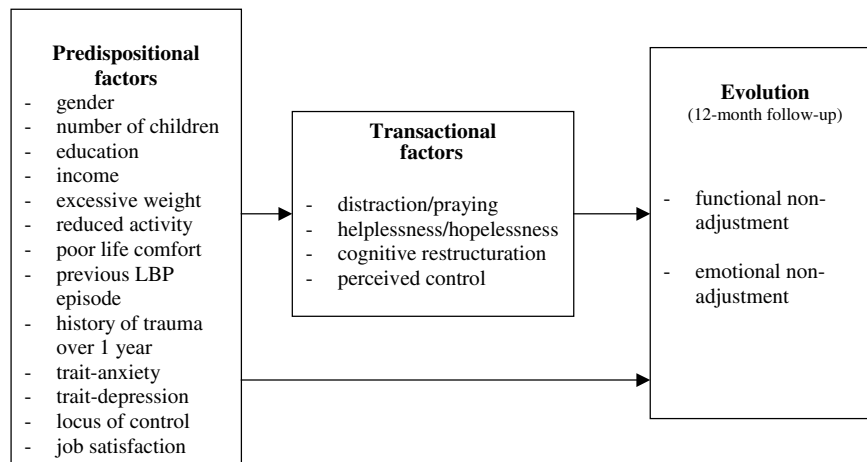


Fig. 1. Analysis model for the hierarchical multiple regressions.

Table 5

Influence of medical, psychological social features and transactional processes on the outcome of LBP

Variables	<i>F</i>	<i>R</i> ²	Adjusted <i>R</i> ²	β
<i>Functional non-adjustment</i>	10.112*	0.473	0.426	
Gender				−0.27*
Number of children				0.15
Education				−0.18
Income				−0.16
History of trauma over one year				0.27*
Reduced activity				0.28*
Poor life comfort				0.09
<i>Emotional non-adjustment</i>	7.225*	0.261	0.225	
Number of children				0.11
Excessive weight				0.16
Previous LBP episode				−0.14
Trait-depression				0.44*
<i>Functional non-adjustment</i>	13.115*	0.236	0.218	
Distraction-praying				0.46*
Perceived control				−0.17
<i>Emotional non-adjustment</i>	14.949*	0.148	0.138	
Helplessness–hopelessness				0.39*

Multiple regression analyses.

* $p < 0.05$.

5. Discussion

Like previous prospective studies, this one emphasized the role of psychosocial factors as powerful predictors of a chronic evolution after LBP (Burton et al., 1995; Fritz et al., 2001; Hasenbring et al., 1994; Klenerman et al., 1995; Linton, 2000). Patients with acute LBP were found to use four main transactional processes to deal with acute LBP, and the role of these four processes in chronic evolution and low quality of life was documented.

However some limitations in the sample collection and methods have to be acknowledged. First, despite participants were enrolled consecutively and prospectively, we found an unusually high rate of chronification: 33% of patients with a poor outcome, persistent pain and low quality of life. This was probably related to the high rate of work-related accidents among the patients with chronic evolution group ($\chi^2 = 8.88$, $p = 0.003$) This outlines the well-known importance of work-related accident and claims in social insurance as risk factors of poor outcome, but prevents to generalize our results to the common population of LBP patients. Second, was our classification system fully accurate? For technical reasons, the interview period was large (10–90 days), and many patients, hoping for a spontaneous recovery without treatment, have been waiting some days before consulting their GP. So we were unable to question the patients just after the onset of pain, and we grouped together in the IP group patients who were very quickly released from pain, others who got relief later in some weeks or months, and six who felt better,

then suffered a relapse, and finally improved. Despite all these 30 participants fulfilled the criteria of being free of pain at one year-follow-up, perhaps their initial coping strategies were different. This has to be looked at in a future study. The same with the recurrent LBP patients, whom we plan to look at in a forthcoming work: although these patients were shown in the literature to take benefit from functional restoration programs as well as non-recurrent LBP patients, Evans et al. demonstrated that they used to develop specific skills, for instance dealing easier with the worker compensation system than the non-recurrent injury patients (Evans et al., 2001; Garcy et al., 1996). So that they might rely on different coping strategies, perception of pain control and social support. Third, our study suffered also from limitations in the assessment battery. Although it is often difficult to interpret data about quality of life in patients with acute pain at the time of the interview, it would have been very interesting to collect MOS and NHP scores at baseline, as to compare with data at T2 time. But our battery was already heavy and of long duration (over one hour), and we did not want to bother our patients further more. As a consequence we could not control for baseline MOS and NHP status in the acute stage.

With regard to the ways these patients reacted to acute LBP, four strategies were identified. The first one, DP, corresponded to a mainly cognitive attitude, which aimed at regulating the emotional content of pain. Patients who used it tried to shift their attention from pain and to think of other things. Patients with

high scores on DP expressed irrational beliefs towards pain. They were thinking that the onset and course of pain were depending on Destiny and/or God's will. They tried to cope with pain by praying, avoiding thinking of pain, and by willingly trying to focus their attention on other things. This strategy may be seen as an adaptive one in acute pain, but seems to play a dysfunctional role on LBP adjustment (Dozois et al., 1996; Rosenstiel and Keefe, 1983; Spinhoven et al., 1989; Riley et al., 1999). In our study, DP was mainly used by men, with external LOC, low trait-anxiety, low income and low job satisfaction, who reduced greatly their activity during acute pain and already had experienced LBP by the past. Patients reacting to pain with the second factor, helplessness–hopelessness, were fully submerged by negative emotions; they were unable to control their feelings of helplessness. They did not believe in cure and recovery. This emotional reaction seems to be specific of dealing with health problems, and has previously been identified in other pathologies (Nuissier, 1996; Tastet, 2001). According to Schermelleh-Engel et al.'s model of pain regulation (1997), the onset of pain makes people overwhelmed with negative emotions. This is in agreement with many previous works emphasizing the well-known role of anxiety and depression in pain perception (Kessler et al., 1996; Lefebvre, 1981; Polatin et al., 1993). Another negative cognition, catastrophizing, has been found associated with a chronic evolution of LBP (Burton et al., 1995; Linton et al., 1999; Picavet et al., 2002). A strong correlation exists between catastrophizing and depression in chronic pain patients (Boothby et al., 1999; Jensen et al., 1994; Rosenstiel and Keefe, 1983; Sullivan et al., 1995). Therefore catastrophizing might be found included in the factor HH. However in our study Catastrophizing, as CSQ-2, was dropped out by the first PCA. So it was impossible for us to look more in detail at its role. We only note that catastrophizing was significantly related to HH ($r = +0.41$, $p < 0.000$). In our study the emotional reaction to pain was mainly used by persons with high levels of trait-anxiety or trait-depression, which seemed logical. These two transactional processes, DP and HH, were found to have a rather negative influence on the outcome.

The third strategy, CR, associated ignorance and reinterpretation of pain sensations. Patients who were using CR tried to give a new, positive meaning to their pain, and to go on with their life as if pain was absent. This strategy associated strong cognitive strategies applied to pain. It has been clearly identified, under different labels, by previous works on LBP coping but its influence on LBP is not clear (Dozois et al., 1996; Rosenstiel and Keefe, 1983; Spinhoven et al., 1989). In our study, it failed to predict emotional and functional adjustment at 12-month follow-up: the use of CR during the acute LBP episode did not seem to influence the evolution of LBP one year later.

The last strategy, PC, was used by people who had feelings of fair and highly available social support and tried to have an active influence on their sensations of pain and tried to control them willingly. This seemed to be a positive attitude, as patients were feeling that they might influence evolution of pain, and had a positive perception of the help they might receive from their family, friends or colleagues. Like Pellino and Oberst (1992) and Haythornthwaite et al. (1998), we observed that PC was mostly used by highly educated persons. It seemed to have some positive influence on the outcome as well: to think that one can control pain during acute LBP improved functional adjustment at 12-month follow-up. However this trend did not reach significance.

With regard now to outcome, we found that chronic evolution could be related either to a functional or to an emotional non-adjustment. FNA involved functional items of quality of life, duration of sick leave, and GP's opinion. ENA logically involved state-anxiety, state-depression, and lack of well-being feelings. This seems very consistent, and allows us to go beyond an over-simple approach of the LBP issue, which would only take into account the subjective feeling of pain, as assessed for instance by visual analog scales.

Medical data, psycho-social features and two transactional processes were found predictive of the two adjustment factors. A great amount of literature has been devoted to demographical and medical predictors of the LBP issue: female gender, excessive weight, job conditions, spinal pathology such as spondylolisthesis or scoliosis have been questioned. Our study provides some news in the field: FNA was predicted by male gender, number of children, low income, low educational level, reduction of activity, poor life comfort, and history of trauma over one year. However male gender, history of trauma over one year, and reduced activity were the unique significant predictors in the present multidimensional model. An interesting point is the link between low activity at T1, during the acute pain, and a chronic evolution of pain: our study confirmed that people who stopped working or moving for a long time did not improve, but on the contrary, experienced more often chronic pain than others. Rest is noxious for the back, as it is now well-known, and it has been at the origin of functional restoration programs (Malmivaara et al., 1995; Mayer et al., 1991).

From a psychological point of view, despite depression and anxiety are well-known to be associated with chronic pain, we did not find higher scores on state-anxiety and depression scales in CLBPP than in IP. However our data fall in agreement with previous works, as they identify trait-depression as the main factor of both functional and emotional low adjustment (Clauw et al., 1999; Epping-Jordan et al., 1998; Pincus et al.,

2002; Polatin et al., 1993). But our study allows to go further, as looking at relationships between these factors and transactional processes. With regard to the influence of these processes, helplessness–hopelessness had a direct negative effect on emotional adjustment, and DP had a direct negative effect on functional one. PC seemed to prevent from a dysfunctional evolution of LBP.

For the future, important issues would be to provide a better understanding of how LBP patients cope with their problem at the end of the acute phase, and to elucidate to what extent prevention issues might be derived from our findings. Indeed chronification of LBP is a complex and multifactorial problem, and multidimensional studies provided evidence that persistent pain would be best predicted by a combination of somatic, psychological and social parameters (Hasenbring et al., 1994). From a therapeutic point of view, all programs relying either on physical treatment only, or on psychological approaches like cognitive-behaviour therapy in isolation failed to prevent acute LBP from becoming a chronic disability problem. Clearly, the spine and the mind are to be treated simultaneously (Polatin et al., 1993). Comprehensive functional restoration treatment programs are now available, which help chronic LBP patients to reduce their disability by enhancing their physical capacities (Brady et al., 1994; Curtis et al., 1994; Mayer et al., 1994) while providing simultaneously psychological support targeted to self-acceptance of pain, training to problem-solving, reducing catastrophizing and fear avoidance (Chaory et al., 2004; Jousset et al., 2004; Khosles et al., 1990). Prevention programs derived from functional restoration concepts are also developing. They include usually identification of risk factors and patients with high-risk, then early active intervention before LBP becomes chronic: fear-avoidance-based physical exercises, problem-solving therapy, and psychological considerations of negative beliefs such as catastrophizing and kinesiophobia (Fritz et al., 2001; George et al., 2003; Picavet et al., 2002). Evidence of their efficacy has been provided (Gatchel et al., 2003; Van den Hout et al., 2003). To what extent identifying ineffective coping attitudes, and trying to make patients to use more accurate coping strategies would add a supplemental value to such programs remains controversial. Some multidisciplinary prevention programs were found to change significantly negative cognitions and coping attitudes, especially PC, along with an improved outcome (Burns et al., 2003; Jensen et al., 2001). In others, the coping strategies demonstrated little to poor change, with a poor prognostic relevance (Pfungsten et al., 1997). In their recent works Kole-Snijders et al. (1999) and Spinhoven et al. (2004) remained doubtful: in comparison with a control group with

passive treatment, cognitive coping skills treatment improved internal pain control and decreased scores for catastrophizing and external pain control as well as operant-behaviour therapy did, but not better: pain coping was not enhanced following treatment. In our study, only helplessness–hopelessness had a clear effect on the emotional issue of low back pain. None of the other transactional processes we looked at had a straight effect on emotional adjustment.

As to conclude, psychosocial factors and transactional processes do play a role when LBP is likely to become a chronic disability. Emotional distress like helplessness–hopelessness and negative beliefs seem to be the most important factors. Future research are needed to provide a better understanding of how acute LBP becomes a chronic disability, and to determine whether early prevention multidisciplinary programs should include psychological intervention on coping strategies.

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References

- Andersson GBJ. Epidemiological features of chronic low back pain. *Lancet* 1999;354:581–5.
- Atlas SJ. The Quebec Task Force classement for the spinal disorders. *Spine* 1996;24:2885–92.
- Bigos SJ, Bowyer OR, Braen GR. Acute low back pain problems in adults. Clinical practise guidelines No. 14. Rockville, MD. Agency for Health Care Policy and Research, December 1994, AHCPR Publication No. 95–0642).
- Boothby JL, Thorn BE, Stroud MW, Jensen MP. Coping with pain. In: Gatchel RJ, Turk DC, editors. *Psychosocial Factors in Pain: Critical Perspectives*. New York: Guilford Press; 1999.
- Brady S, Mayer T, Gatchel RJ. Physical progress and residual impairment quantification after functional restoration. Part II: Isokinetic trunk strength. *Spine* 1994;19(4):395–400.
- Bruchon-Schweitzer M, Paulhan I. L'inventaire d'anxiété-trait et d'anxiété-état. Adaptation française et validation du S.T.A.I.-Yde C.D. Spielberger. Paris: Les Editions du Centre de Psychologie Appliquée; 1993.
- Bucquet D, Condon S, Ritchie K. The French version of the Nottingham Health Profile. A comparison of items weights with those of the source version. *Soc Sci Med* 1990;30(7):829–35.
- Burns JW, Kubilus A, Bruehl S, Harden RN, Lofland K. Do changes in cognitive factors influence outcome following multidisciplinary treatment for chronic pain? A cross-lagged panel analysis. *J Consult Clin Psychol* 2003;71(1):81–91.
- Burton AK, Tillotson KM, Main CJ, Hollis S. Psychosocial predictors of outcome in acute and subchronic low back trouble. *Spine* 1995;20(6):722–8.

- Cattell RB. *Handbook of Multivariate Experimental Psychology*. Chicago: Rand McNally; 1966.
- Chaory K, Rannou F, Fermanian J, Genty M, Rosenberg S, Billaber G, et al. Impact of functional restoration programs on fears, avoidance and beliefs in chronic low back pain patients. *Ann Readapt Med Phys* 2004;47(3):93–7.
- Clauw DJ, Williams D, Lauerma W, Dahlman M, Aslami A, Nachemson AL, et al. Pain sensitivity as a correlate of clinical status in individuals with chronic low back pain. *Spine* 1999;24(19):2036–41.
- Clinical guidelines for the management of acute low back pain. London: Royal College of General Practitioners, 1996.
- Curtis L, Mayer T, Gatchel RJ. Physical progress and residual impairment quantification after functional restoration. Part III: Isokinetic and isoinertial lift capacity. *Spine* 1994;19(4):401–5.
- Dozois DJA, Dobson KS, Wong M, Hughes D, Long A. Predictive utility of the CSQ in low back pain: individual vs composite measures. *Pain* 1996;66:171–80.
- Duquesnoy B. Définition de la lombalgie chronique. *Rev Rhum* 1994;61(4bis):09s–10s.
- Epping-Jordan JE, Wallgren DR, Williams RA, Pruitt SD, Slater MA, Patterson TL, et al. Transition to chronic pain in men with low back pain: predictive relationships among pain intensity, disability, and depressive symptoms. *Health Psychol* 1998;17(5):421–7.
- Evans TH, Mayer TG, Gatchel RJ. Recurrent disabling work-related spinal disorders after prior injury claims in a chronic low back pain population. *Spine* 2001;26(1):183–9.
- Fabrigar LR, Wegener DT, MacCallum RC, Strahan EJ. Evaluating the use of exploratory factor analysis in psychological research. *Psychol Meth* 1999;4(3):272–99.
- Fisher K, Johnston M. Emotional distress and control cognitions as mediators of the impact of chronic pain on disability. *Br J Health Psychol* 1998;3:225–36.
- Fritz JM, George SZ, Delitto A. The role of fear-avoidance beliefs in acute low back pain: relationships with current and future disability and work status. *Pain* 2001;94(1):7–15.
- Fuhrer R, Rouillon F. La version française de l'échelle CES-D, Description et traduction de l'échelle d'auto-évaluation. *Psychiatrie Psychobiol* 1989;4:163–6.
- Garcy P, Mayer TG, Gatchel RJ. Recurrent or new injury outcomes after return to work in chronic disabling disorders. Tertiary prevention efficacy of functional restoration program. *Spine* 1996;21(8):952–9.
- Gatchel RJ, Polatin PB, Mayer TG. The dominant role of psychosocial risk factors in the development of chronic low back pain disability. *Spine* 1995;20(24):2702–9.
- Gatchel RJ, Polatin PB, Noe C, Gardea M, Pulliam C, Thompson J. Treatment and cost-effectiveness of early intervention for low-back pain patients: a one-year prospective study. *J Occup Rehabil* 2003;13(1):1–9.
- George SZ, Fritz JM, Bialosky JE, Donald DA. The effect of a fear-avoidance-based physical therapy intervention for patients with acute low back pain: results of a randomized trial. *Spine* 2003;28(23):2551–60.
- Guidelines for the management of back-injured employees. Melbourne, Australia: Workcover Corporation, October 1993. p. 1–44.
- Hasenbring M, Marienfeld G, Kuhlendahl D, Soyka D. Risk factors of chronicity in lumbar disc patients. A prospective investigation of biologic, psychologic and social predictors of therapy outcome. *Spine* 1994;19(24):2759–65.
- Haythornthwaite JA, Menefee LA, Heinberg LJ, Clark MR. Pain coping strategies predict perceived control over pain. *Pain* 1998;77:33–9.
- Hazard RG, Haugh LD, Reid S, Preble JB, MacDonald L. Early prediction of chronic disability after occupational low back injury. *Spine* 1996;21:945–51.
- Hunt SM, Wilkund I. Cross-cultural variation in the weighting of health statements: a comparison of English and Swedish valuations. *Health Policy* 1987;8:230.
- Jensen MP, Turner JA, Romano JM. Correlates of improvement in multidisciplinary treatment of chronic pain. *J Consult Clin Psychol* 1994;62(1):172–9.
- Jensen MP, Turner JA, Romano JM. Changes in beliefs, catastrophizing, and coping are associated with improvement in multidisciplinary pain treatment. *J Consult Clin Psychol* 2001;69(4):655–62.
- Jousset N, Fanello S, Bontoux L, Dubus V, Billabert C, Vielle B, et al. Effects of functional restoration versus 3 hours per week of physical therapy: a randomized controlled study. *Spine* 2004;29(5):487–93.
- Kessler M, Kronstorfer R, Traue HC. Depressive symptoms and disability in acute and chronic back pain patients. *Int J Behav Med* 1996;3(2):91–103.
- Kholes S, Barnes D, Gatchel RJ, Mayer TG. Improved physical performance outcomes after functional restoration treatment in patients with chronic low-back pain. Early versus recent training results. *Spine* 1990;15(12):1321–4.
- Klenerman L, Slade PD, Stanley IM, Pennie B, Reilly JP, Atchison LE, et al. The prediction of chronicity in patients with an acute attack of low back pain in a general practice setting. *Spine* 1995;20(4):478–84.
- Koleck M. Rôle de certains facteurs psychosociaux dans le profil évolutif des lombalgies communes. In: Schweitzer M, Quintard B, editors. *Bruchon- Personnalité et maladies*. Paris: Dunod; 2001.
- Kole-Snijders AM, Vlaeyen JW, Goossens ME, Rutten-van Molken, Heuts PH, van Breukelen G. Chronic low-back pain: what does cognitive coping skills training add to operant behavioral treatment? Results of a randomized clinical trial. *J Consult Clin Psychol* 1999;67(6):931–44.
- Kröner-Herwig B, Jäkle C, Frettlöh J, Peters K, Seemann H, Franz C. Predicting subjective disability in chronic pain patients. *Int J Behav Med* 1996;31(1):30–41.
- Lefebvre MF. Cognitive distortion and cognitive errors in depressed psychiatric and low back pain patients. *J Consult Clin Psychol* 1981;49(4):517–25.
- Lewandowski W. Psychological factors in chronic pain: a worthwhile undertaking for nursing. *Arch Psychiatr Nurs* 2004;18(3):97–105.
- Lin CC, Ward SE. Perceived self-efficacy and outcome expectancies in coping with chronic low back pain. *Res Nurs Health* 1996;19:299–310.
- Linton SJ. A review of psychological risk factors in back and neck pain. *Spine* 2000;25(9):1148–56.
- Linton SJ, Hallden K. Can we screen for problematic back pain? A screening questionnaire for predicting outcome in acute and subacute back pain. *Clin J Pain* 1998;14:209–15.
- Linton SJ, Buer N, Vlayen J, Hellsing AL. Are fear-avoidance beliefs related to a new episode of back pain? A prospective study. *Psychol Health* 1999;14:1051–9.
- Lumpkin JR. Validity of a brief Locus of Control Scale for survey research. *Psychol Rep* 1985;57:655–9.
- Maetzel A, Li L. The economic burden of low back pain: a review of studies published between 1996 and 2001. *Best Pract Res Clin Rheumatol* 2002;16(1):23–30.
- Malmivaara A, Hakkinen U, Aro T, Heinrichs ML, Koskeniemi L, Kuosma E, et al. The treatment of acute low back pain: bed rest, exercises, or ordinary activities? *N Eng J Med* 1995;332(6):351–5.
- Mayer T, Mooney V, Gatchel R. *Contemporary conservative care for painful spinal disorders*. Philadelphia: Lea & Fibiger; 1991.
- Mayer T, Tabor J, Bovasso E, Gatchel RJ. Physical progress and residual impairment quantification after functional restoration. Part I: Lumbar mobility. *Spine* 1994;19(4):389–94.
- McCracken LM, Goetsch VL, Semenchuk EM. Coping with pain produced by physical activity in persons with chronic low back pain: immediate assessment following a specific pain event. *Behav Med* 1998;24:29–34.

- Nuissier J. Les facteurs de l'ajustement au traumatisme médullaire: une étude en psychologie de la santé. Thèse de doctorat, Université Bordeaux II, 1996.
- Pellino TA, Oberst MT. Perception of control and appraisal of illness in chronic low back pain. *Orthop Nurs* 1992;11(1):22–6.
- Pfingsten M, Hildebrandt J, Leibing E, Franz C, Saur P. Effectiveness of a multimodal treatment program for chronic low-back pain. *Pain* 1997;73(1):77–85.
- Picavet HS, Vlaeyen JW, Schouten JS. Pain catastrophizing and kinesiophobia: predictors of chronic low back pain. *Am J Epidemiol* 2002;56(11):1028–34.
- Pincus T, Burton AK, Vogel S, Field AP. A systematic review of psychosocial factors as predictors of chronicity/disability in prospective cohorts of low back pain. *Spine* 2002;27(5):109–20.
- Polatin PB, Kinney RK, Gatchel RJ, Lillo E, Mayer TG. Psychiatric illness and chronic low back pain. The mind and the spine – which goes first. *Spine* 1993;18(1):66–71.
- Pope MH. Risk indicators in low back pain. *Ann Med* 1989;21:387–92.
- Potter RG, Jones JM. The evolution of chronic pain among patients with musculoskeletal problems: a pilot study in primary care. *Br J Gen Pract* 1992;42:462–4.
- Pruyn JFA, Van Der Born HW, De Reuver RSM, De Boer MF, Bosnian LJ, Ter Pelkewijk MA. Locus of Control-Schaal voor kankerpatienten. (The Locus of Control Scale for cancer patients). *Tijdschrift voor Sociale Gezondheidszorg* 1988;66:404–8.
- Radloff LS. The CES-D Scale: a self-report depression scale for research in the general population. *Appl Psychol Measure* 1977;1:385–401.
- Riley JL, Robinson ME, Geisser ME. Empirical subgroups of the Coping Strategies Questionnaire-Revised: a multisample study. *Clin J Pain* 1999;5:111–6.
- Robinson ME, Riley JL, Myers CD, Sadler IJ, Kvaal SA, Geisser ME, et al. The Coping Strategies Questionnaire: a large sample, item level factor analysis. *Clin J Pain* 1997;13:43–9.
- Rosenstiel AK, Keefe FJ. The use of coping strategies in chronic low back pain patients: relationship to patient characteristics and current adjustment. *Pain* 1983;17:33–44.
- Sarason IG, Levine HM, Basham RB, Sarason BR. Assessing social support: the social support questionnaire. *J Pers Soc Psychol* 1983;44:127–39.
- Schermelleh-Engel K, Eifert GH, Moosbrugger H, Frank D. Perceived competence and trait anxiety as determinants of pain coping strategies. *Person Individ Diff* 1997;22(1):1–10.
- Spielberger CD, Goruch RL, Lushene RE, Vagg PR, Jacobs GA. Manual for the State-Trait Inventory (STAI) Form Y. Palo Alto: Consulting Psychologists Press; 1983.
- Spinhoven P, Ter Kuile MM, Linssen ACG, Gazendam B. Pain coping strategies in a Dutch population of chronic low back pain patients. *Pain* 1989;45:29–34.
- Spinhoven P, Ter Kuile M, Kole-Snijders AM, Hutten Mansfeld M, Ouden DJ, Vlaeyen JW. Catastrophizing and internal pain control as mediators of outcome in the multidisciplinary treatment of chronic low back pain. *Eur J Pain* 2004;8(3):211–9.
- Stewart AL, Hays RD, Ware JE. The MOS Short-Form General Health Survey, reliability and validity in a patient population. *Med Care* 1988;26(7):724–34.
- Sullivan MJL, Bishop SR, Pivik J. The pain catastrophizing scale: development and validation. *Psychol Assess* 1995;7(4):524–32.
- Symonds TL, Burton AK, Tillotson KM, Main CJ. Do attitudes and beliefs influence work loss due to low back trouble. *Occup Med* 1996;46(1):25–32.
- Tastet S. Approche biopsychosociale des cancers du sein : stratégies d'ajustement et immunocompétence. Thèse de doctorat, Université Bordeaux II, 2001.
- Truchon M, Fillion L. Biopsychosocial determinants of chronic disability and low-back pain: a review. *J Occup Rehabil* 2000;10(2):117–42.
- Tuttle DH, Shuttly MS, Degood DE. Empirical dimensions of coping in chronic pain patients: a factorial analysis. *Rehabilitat Psychol* 1991;36(3):179–88.
- Välfors B. Acute, subacute and chronic low back pain: clinical symptoms, absenteeism and working environment. *Scand J Rehab Med* 1985;11:1–98.
- Van den Hout JH, Vlaeyen JW, Heuts PH, Zijlema JH, Wijnen JA. Secondary prevention of work-related disability in nonspecific low back pain: does problem-solving therapy help? A randomized clinical trial. *Clin J Pain* 2003;19(2):87–96.
- Ware JE, Sherbourne CD. The MOS 36-item Short-Form Health Survey (SF 36) I -conceptual framework and item selection. *Med Care* 1992;30(6):473–83.